08 – Hallucinations & fact-checking

Definition: A **hallucination** is a made-up but convincing-sounding answer from the AI, for example a wrong fact, a non-existent source, or an invented section of law.

> **Remember:** The more consequential the answer, the more thoroughly you check.

Key points

- **Why does this happen?** An LLM predicts likely text fragments (topic 01). When it lacks solid knowledge, it fills the gap with something plausible instead of saying “I don’t know”.
- **Typical risk areas:** numbers, studies, quotes, legal matters (sections of law, deadlines), medicine, information about people, links and sources, as well as code packages that don’t exist at all.
- **Warning signs:** very precise details without evidence, sources without a link, smooth answers to niche questions. But style never reliably tells you whether something is true.
- **Remedies:** Use web search or ask for sources and then **open them yourself** (links can be invented too). Allow uncertainty. Provide your own text (topic 07). Ask a second source or a second model. When in doubt, read the original source.
- **Even with web search**, the AI can misrepresent sources. Check whether the claim really is there.
- **“Are you sure?” is no proof.** The confirmation can be invented too.
- **How deeply to check (rule of thumb):** Check ideas and drafts briefly. For money, health, legal matters, and publication, always check against the original source.

Example (before/after)

You need references for a talk.

Before (risky):

```
Name three studies that prove that breaks improve concentration.
```

The AI can invent titles, authors, and years that look real.

After:

```
Summarize what is known about the topic "breaks and concentration".
Name sources only if you are sure of them. Otherwise write "not sure".
I will look up and check every source myself.
```

Do / Don’t

Do	Don’t
Open and read sources yourself	Copy source references without checking
Explicitly allow “I don’t know”	Insist on an answer the AI doesn’t have

Do	Don’t
Check with varying thoroughness depending on the consequences	Pass on AI texts unchecked

Safety note

For health and legal matters, the AI is no substitute for a professional. Use it to prepare, but don’t decide on that basis alone.

Mini checklist

- ☐ I can explain what a hallucination is.
- ☐ I know the typical risk areas.
- ☐ I open sources myself instead of just taking them over.
- ☐ I match the depth of checking to the consequences.

Exercise (5 min.)

Ask the AI to name three sources on an everyday topic. Check each source: Does it exist? Does the link open? Is the claim really stated there? Note how many sources passed the check.

Next

- Extra exercises: `02_extra_exercises/08_extra_exercises`
- Test: Test B (Good results)
- Related: 01 What is AI?, 07 Prior knowledge & context, 11 AI tools

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